UMA069: GRAPH THEORY AND APPLICATIONS

L	T	P	Cr
2	0	0	2.0

Course Objective: The objective of the course is to introduce students with the fundamental concepts in graph Theory, with a sense of some its modern applications. They will be able to use these methods in subsequent courses in the computer, electrical and other engineering.

Syllabus

Introduction: Graph, Finite and infinite graph, incidence and degree, Isolated vertex, Pendent vertex and null graph, Isomorphism, Sub graph, Walks, Paths and circuits, Euler circuit and path, Hamilton path and circuit, Euler formula, Homeomorphic graph, Bipartite graph, Edge connectivity, Computer representation of graph, Digraph.
Tree and Fundamental Circuits: Tree, Distance and center in a tree, Binary tree, Spanning tree, Finding all spanning tree of a graph, Minimum spanning tree.
Graph and Tree Algorithms: Shortest path algorithms, Shortest path between all pairs of vertices, Depth first search and breadth first of a graph, Huffman coding, Cuts set and cut vertices, Warshall's algorithm, topological sorting.
Planar and Dual Graph: Planner graph, Kuratowski's theorem, Representation of planar graph, five-color theorem, Geometric dual.
Coloring of Graphs: Chromatic number, Vertex coloring, Edge coloring, Chromatic partitioning, Chromatic polynomial, covering.
Application of Graphs and Trees: Konigsberg bridge problem, Utilities problem, Electrical network problem, Seating problem, Chinese postman problem, Shortest path problem, Job sequence problem, Travelling salesman problem, Ranking the participant in a tournament, Graph in switching and coding theory, Time table and exam scheduling, Applications of tree and graph in computer science.

Laboratory Work: NA

Course Learning Objectives (CLO)

Upon completion of the course, the students will be able to:

1. understand the basic concepts of graphs, directed graphs, and weighted graphs and able to present a graph by matrices.
2. understand the properties of trees and able to find a minimal spanning tree for a given weighted graph.
3. understand Eulerian and Hamiltonian graphs.
4. apply shortest path algorithm to solve Chinese Postman Problem.
5. apply the knowledge of graphs to solve the real-life problem.

Recommended Books:

1. Deo, N., Graph Theory with Application to Engineering with Computer Science, PHI, New Delhi (2007)
2. West, D. B., Introduction to Graph Theory, Pearson Education, London (2008)
3. Bondy, J. A. and Murty, U.S.R., Graph Theory with Applications, North Holland Publication, London (2000)
4. Rosen, K. H., Discrete Mathematics and its Applications, Tata-McGraw Hill, New Delhi (2007)